

Quantifying life history

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Learning objectives

1. Understand how and why growth, survival, and fecundity trade off
2. Do sensitivity and elasticity analyses

review of last lecture: life tables and structured populations

in Silvertown discussion, talked about where species fall in demographic triangle

this week: talk a bit more about plant strategies & analyzing structured populations by manipulating elements of the matrix (vital rates)

Tradeoffs

you can't maximize growth, survival, and reproduction

limited amount of energy and resources - need to decide how to allocate

leads to common life history strategies, tradeoffs between vital rates

most basic tradeoff is the r - K theory

r - K

- 2 strategies, can be a spectrum
- r species - fast intrinsic growth r , many offspring, low investment
- K species - slow growth, near carrying capacity K , few offspring, high investment

question: what environments might favor a species closer to r vs. K ?

- **caveat** - oversimplified example, trees like oaks that are long-lived but produce loads of acorns with low investment per acorn defy this theory

Demographic triangle

- 3 strategies
- growth, stasis, fecundity

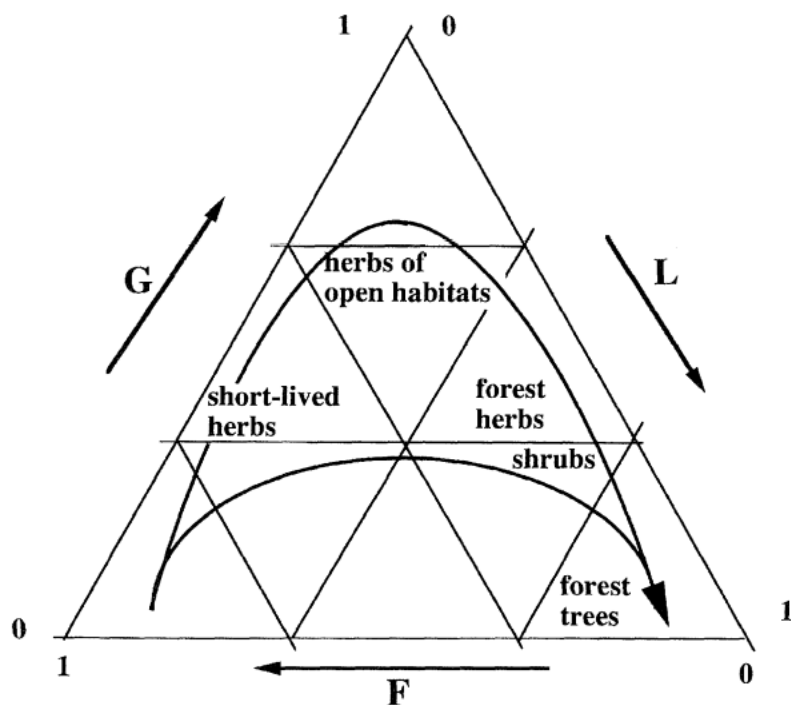


Figure 2. The successional trajectories suggested by the distribution of species in the demographic triangle in Fig. 1, after Silvertown & Franco (1993).

Sensitivity analyses

Sensitivity analysis tells you how much the population growth rate would increase/decrease if you increased/decreased a vital rate

- absolute change in λ with change in vital rate
- perturb specific stage/age *e.g.* increase survival of adults with grazing exclosure
- repeatedly compute population growth rate

Demonstration

we're going to perturb adult survival

```
# population matrix model from last time
stage <- c(0,0.2,0,0,0.1,0.3,4,0,0.8)
stage_mat <- matrix(stage,nrow=3,ncol=3)
stage_mat
```

```
##      [,1] [,2] [,3]
## [1,]  0.0  0.0  4.0
## [2,]  0.2  0.1  0.0
## [3,]  0.0  0.3  0.8
```

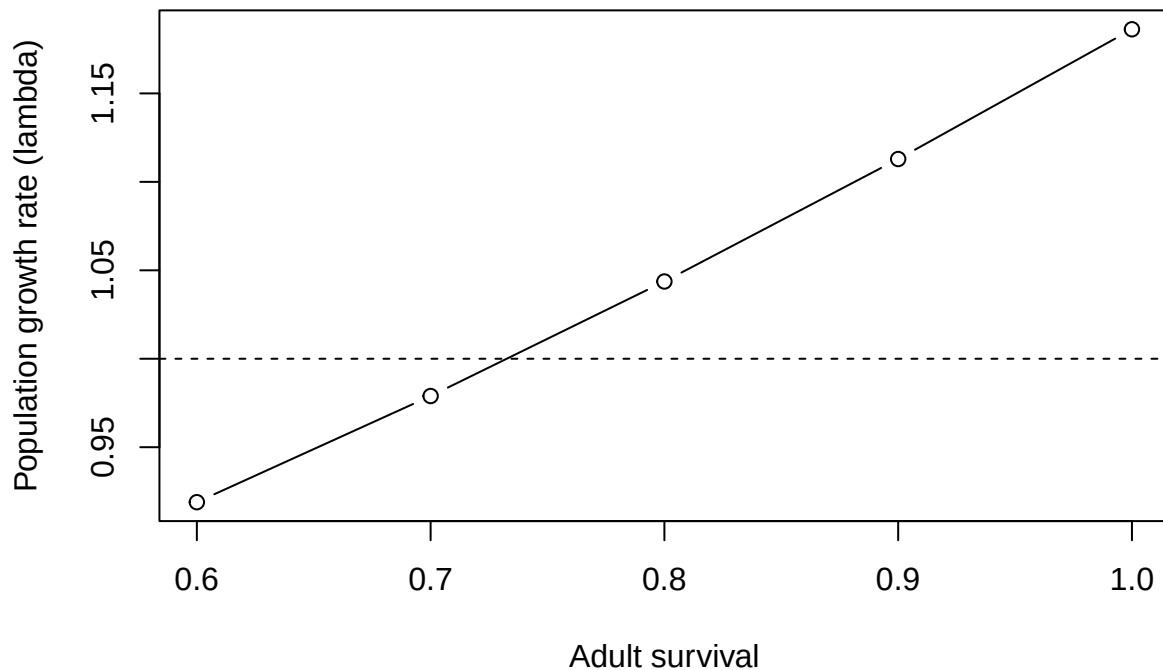
```
# new adult survival values (instead of 0.8)
adult_survival <- seq(0.6, 1.0, by = 0.1)
```

```
# get population growth rates from range of adult survival values
lambda_values <- sapply(adult_survival, function(x) {
  pert_mat <- stage_mat
  pert_mat[3, 3] <- x
  max(Re(eigen(pert_mat)$values))
})
```

```
lambda_values
```

```
## [1] 0.9189244 0.9789340 1.0436797 1.1129045 1.1862529
```

```
# plot
plot(adult_survival, lambda_values,
     type = "b",
     xlab = "Adult survival",
     ylab = "Population growth rate (lambda)",
     abline(h = 1, lty = 2))
```



Elasticity analyses

let's say you have rare plant population data. Population growth rate is less than 1, declining, and you're interested in figuring out what vital rate is most important for population growth

- quantify influence of vital rates on λ
- relative importance/contribution of vital rates
- proportional/multiplicative change in growth rate

Demonstration

```
stage_mat
```

```
##      [,1] [,2] [,3]
## [1,] 0.0  0.0  4.0
```

```
## [2,] 0.2 0.1 0.0
## [3,] 0.0 0.3 0.8

elasticity <- matrix(NA, nrow = 3, ncol = 3)

lambda <- max(Re(eigen(stage_mat)$values))

for (i in 1:nrow(stage_mat)) {
  for (j in 1:ncol(stage_mat)) {

    pert <- 0.01

    if(i == 1 & j == 1) print(pert)

    # take our original matrix, add the change to each component, IN TURN (not all at once)
    elas_mat <- stage_mat
    elas_mat[i, j] <- elas_mat[i, j] + pert

    if(i == 1 & j == 1) print(elas_mat)

    # get the PROPORTIONAL change in population growth rate (divide by the change in vital rate/perturbation)
    sensitivity <- (max(Re(eigen(elas_mat)$values)) - lambda) / pert

    if(i == 1 & j == 1) print(sensitivity)

    # computation for elasticity: have the proportional lambda change, scale by the original proportion
    elasticity[i, j] <- (stage_mat[i, j] / lambda) * sensitivity
    # 0/lambda for any elements with 0
  }
}
```

```
## [1] 0.01
##      [,1] [,2] [,3]
## [1,] 0.01 0.0 4.0
## [2,] 0.20 0.1 0.0
## [3,] 0.00 0.3 0.8
## [1] 0.1574172
```

```
elasticity
```

```
##      [,1]      [,2]      [,3]
## [1,] 0.0000000 0.0000000 0.1564229
## [2,] 0.1546185 0.01668837 0.0000000
## [3,] 0.0000000 0.15524354 0.5155701
```

question: if you have a population of plants and you supplemented fecundity by sowing seeds (or reduced survival with herbicide) then measured population growth for the next few years, is that a sensitivity or elasticity analysis?